

Dr. Prasanta Majee

M. Tech, Ph.D.

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Profile

Researcher with a Ph.D in Civil Engineering, specializing in the behavior of organic contaminants and heavy metal removal, utilizing advanced analytical techniques. Expertise in pesticide sorption, degradation, leaching, soil remediation, and water quality assessment, with hands-on experience in high-performance liquid Chromatography (HPLC), Gas Chromatography (GC), and UV-visible spectroscopy. Proficient in water and wastewater treatment, solid waste management, and biochar-based adsorption technologies. Demonstrated ability to tackle complex environmental challenges with innovative solutions, emphasizing sustainability and interdisciplinary collaboration.

Education

2018 – 2024 Warangal, India	Doctor of Philosophy (PhD) in Civil Engineering <i>National Institute of Technology Warangal (NIT Warangal)</i> Title of the Thesis: "Behavior of Pesticides in Agricultural Soils: Assessment and Biochar-Based Control Approach"
2015 – 2017 Warangal, India	M.Tech in Environmental Engineering, <i>National Institute of Technology Warangal (NIT Warangal)</i> Grade: First Class, 7.38/10-point scale Title of the Thesis: "Spatial and Temporal Variation of Groundwater Quality in Warangal City"
2010 – 2014 Haldia, India	B.Tech in Civil Engineering <i>Haldia Institute of Technology</i> Grade: First Class, 8.14/10-point scale Project Work: "Design of 12-Storied Office Building"

Professional Experience

01/2025 – present Warangal, India	Senior Project Scientist <i>Department of Civil Engineering, National Institute of Technology Warangal (NIT Warangal)</i> • Leading NRCO-sponsored environmental assessment and management planning for the Krishna River Basin • Designing and executing field investigations and sampling campaigns for surface water, groundwater, and pollution source identification • Conducting integrated assessment of industrial wastewater and solid waste impacts on river systems • Preparing technical reports and basin management plans aligned with regulatory and sustainability frameworks • Supporting environmental decision-making through data-driven analysis and risk assessment • Coordinating with multidisciplinary stakeholders for project execution
2018 – 2024 Warangal, India	Doctoral Researcher (Ph.D. Scholar) <i>National Institute of Technology, Warangal (NIT Warangal)</i> • Conducted comprehensive assessment of pesticide contamination and its transport in soil-groundwater systems • Evaluated environmental risks associated with contaminant leaching and groundwater pollution • Designed laboratory and field experiments to simulate real-world contamination scenarios • Developed remediation strategies using biochar to reduce contaminant mobility • Performed chemical analysis using HPLC and GC to support environmental monitoring • Contributed to water and soil pollution control frameworks supporting sustainable agricultural practices.
06/2016 – 07/2016 Visakhapatnam, India	Intern <i>Hindustan Petroleum Corporation Limited (HPCL)</i> • Assisted in environmental monitoring and evaluation of industrial wastewater treatment systems • Supported compliance reporting aligned with environmental regulations • Evaluated treatment efficiency and operational performance of ETP systems

Research Experience

2018 – 2024

PhD Research Work

Thesis Title: "Behavior of Pesticides in Agricultural Soils: Assessment and Biochar-Based Control Approach"

Pesticides have become an indispensable component of modern agricultural practices, gaining widespread popularity among farmers as they function as protective shields for crops. Over the past five years, there has been a substantial five-fold increase in pesticide application in the state of Telangana, raising concerns about the preservation of groundwater and surface water resources. To address these concerns, a study was conducted to investigate the fate and behaviour of two widely used pesticides, Imidacloprid and Atrazine, in four different soils within the region. The research aimed to assess their potential to contaminate groundwater using various methodologies. The study revealed that these pesticides are prone to contaminating water sources. In response to these findings, various biochars and hydrochars were prepared as effective measures to control the leaching of pesticides in the soil, and this goal was successfully achieved. The biochars demonstrated the ability to control the mobility of pesticides in soils. This research underscores the importance of implementing strategies to mitigate the environmental impact of pesticide use and emphasises the potential role of biochars in promoting sustainable agricultural practices.

2016 – 2017

M.Tech Research Work

Thesis Title: "Spatial and Temporal Variation of Groundwater Quality in Warangal City"

Groundwater quality was assessed at a total of 23 sample locations over a six-month period. Twelve physicochemical parameters and six different heavy metals were investigated across all samples. Water Quality Index (WQI) was developed for each location to assess its suitability for drinking purposes. The entire region was mapped using ArcGIS to visualize the spatial variation of each parameter, providing insight into the overall quality of the study area. The findings of the study proved valuable to local government authorities for planning water supply initiatives in the area.

Technical Skills

Environmental Engineering:

- Environmental Site Assessment (Phase I & II ESA)
- Soil & Groundwater Contamination Assessment
- Environmental Impact Assessment (EIA)
- Risk Assessment & Regulatory Compliance
- Wastewater Treatment Systems
- Environmental Monitoring & Site Investigation

Analytical & Laboratory:

- HPLC, Gas Chromatography (GC), UV-Vis Spectroscopy
- Organic Contaminant & Heavy Metal Analysis
- Soil & Water Sampling Techniques

Tools & Data:

- ArcGIS, EPANET
- R (Basic), MS Excel
- Data Analysis & Interpretation

Languages

- English
- Bengali
- Hindi

Publications

- **Prasanta Majee** & P Hari Prasad Reddy (2023), "Sorption behaviour of atrazine on agricultural soils of different characteristics: equilibrium and kinetics studies" **Clean Technologies and Environmental Policy**, 25,3407-3417 (SCI, I.F-4.3)
<https://doi.org/10.1007/s10098-023-02600-5> 
- **Prasanta Majee** & P Hari Prasad Reddy (2023), "Sorption desorption behaviour of Imidacloprid in agricultural soils of Deccan plateau of Southern India: kinetics and thermodynamic analysis", **International Journal of Environmental Analytical Chemistry**, 104(20), 8570–8584 (SCI, I.F-2.6)
<https://doi.org/10.1080/03067319.2023.2206964> 
- **Prasanta Majee**, Sudheekar Reddy Periyavaram, Lavakumar Uppala & P Hari Prasad Reddy (2025), "Physicochemical and Energy Characteristics of Biochar and Hydrochar derived from Cotton Stalks: A Comparative Study", **BioEnergy Research**, 18, 20 (SCI, I.F- 3.1)
<https://doi.org/10.1007/s12155-025-10816-0> 
- Samar Mahata, **Prasanta Majee**, Abhisek Mondal & Brajesh Kumar Dubey (2026), "Decoding Valorization of Hydrothermal Carbonization Process Water: From Chemical Complexity to Circular Waste Management Pathways". **ACS ES&T Engineering**, (Accepted)

- Samar Mahata, **Prasanta Majee**, Sudheekar Reddy Periyavaram, & M Chandrasekhar (2025), "Optimization of energy recovery through the catalytic Co-Hydrothermal Carbonization of sewage sludge and lignocellulose biomass", **Biomass Conversion and Biorefinery, (SCI, I.F-3.5)**
<https://doi.org/10.1007/s13399-025-06874-1> 
- Lavakumar Uppala, Sivaprakash S, **Prasanta Majee**, Sudheekar Reddy Periyavaram, P Hari Prasada Reddy, Shirish Hari Sonawane (2025), "Sustainable Hydrothermal Carbonization of Yard Waste: Thermal Behaviour and Adsorption Studies Using Distilled Water and Leachate", **Journal of Hazardous, Toxic, and Radioactive Waste Management**, 30 (1), 04025033 (ASCE)
<https://doi.org/10.1061/JHTRBP.HZENG-1541> 
- Gauri Shankar Gupta, **Prasanta Majee**, Lavakumar Uppala, Sudheekar Reddy Periyavaram, P Hari Prasada Reddy (2025), " Valorization of Flower Waste: Natural Dye Extraction and Hydrochar Production". **Engineering Research Express**, 7(4), 045147 (SCOPUS)
10.1088/2631-8695/ae2cea
- Ankita Hazarika, Shubham Giri, **Prasanta Majee**, P Hari Prasad Reddy (2025), "Removal of Microplastics from Water by Using Agro-Waste Biochar". Proceedings of 9IYGEC 2023, Volume 2. IYGEC 2021. Lecture Notes in Civil Engineering, vol 606. Springer, Singapore.
https://doi.org/10.1007/978-981-97-6988-9_26 
- Suraj Bohra, **Prasanta Majee**, Lavakumar Uppala, P Hari Prasada Reddy, (2026), "Nanoparticle-Assisted Photocatalytic Degradation of Rhodamine B Dye: A Sustainable Approach to Wastewater Treatment". Lecture Notes in Civil Engineering, vol 773. Springer, Singapore.
https://doi.org/10.1007/978-981-95-3775-4_12 
- **Prasanta Majee**, Shubham Giri & P Hari Prasad Reddy (2025), "Efficacy of Cotton Stalk Biochar on Sorption Behavior and Leaching Potential of Imidacloprid in Agricultural Soils: A Comprehensive Study", **Journal of Hazardous, Toxic, and Radioactive Waste Management, (ASCE) (Under Review)**
- **Prasanta Majee**, Shubham Giri & P Hari Prasad Reddy (2025), "Advancing Understanding of Pesticide Sorption Dynamics in Agroecosystems: Role of Soil Amendments and Environmental Interactions", **International Journal of Environmental Analytical Chemistry, (SCI) (Under Review)**

Conferences

- **ASCE International Conference on Challenges and Innovations for Sustainable Smart Cities (CISSC 2025)**, 7th -9th February 2025, Chandigarh, India
- **5th Euro- Mediterranean Conference for Environmental Integration**, 2nd -5th October 2023, Rende, Italy
- **Ninth Indian Young Geotechnical Engineers Conference (9IYGEC)**, Aurangabad, India, 21st-22nd March, 2023
- **International Conference on Innovative Trends in Civil Engineering for Sustainable Development (ITCSD-2019)**, NIT Warangal, 13th-15th September, 2019.

References

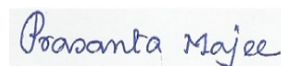
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Declaration

I hereby declare that the information provided by me above is true to the best of my knowledge and belief.



Dr. Prasanta Majee